Wenrui Jiang

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Education

Johns Hopkins University *Sept 2021 – Dec 2025*

- **Department of Earth and Planetary Sciences**
- **Institute for Data Intensive Engineering and Science**
- **Department of Mechanical Engineering**

Ph.D. in Physical Oceanography (2025)
M.S. in Earth and Planetary Sciences (2025)
M.S.E. in Mechanical Engineering (2024)

Fudan University *Sept 2017 - June 2021*

- **Department of Atmospheric and Oceanic Sciences**

B.S. in Atmospheric Science (2021)
Minor in Data Science (2021)

UCLA *Sept 2019 - Dec 2019*

- **Department of Atmospheric and Oceanic Sciences**

Exchange student at Department of Atmospheric and Oceanic Sciences

Professional Experience

University of Cambridge *Oct 2026 – present*

- **Department of Applied Mathematics & Theoretical Physics**

Research Associate

Massachusetts Institute of Technology *Jan 2026 – Aug 2026*

- **Department of Earth, Atmospheric & Planetary Sciences**

Postdoctoral Associate

Publications

[1] **Wenrui Jiang**, Thomas W. N. Haine, and Peter P. Sullivan. Dynamics of monochromatic inertia–gravity waves crossing an upper-ocean submesoscale cold filament. *(In Prep)*, 2026.

[2] **Wenrui Jiang**, Peter P. Sullivan, and Thomas W. N. Haine. Energetics of submesoscale frontogenesis and arrest in an upper-ocean dense filament. *J. Fluid Mech. (Under Review)*, 2026.

[3] Carenza M. F. Williams, **Wenrui Jiang**, and Thomas W. N. Haine. Two phase freshening of the Beaufort gyre between 2003 and 2016. *J. Geophys. Res. Oceans(Under review)*, 2026.

[4] Yilang Xu, Renske Gelderloo, **Wenrui Jiang**, and Thomas W. N. Haine. Influence of sub-inertial variability on dense overflows across the Iceland-Faroe Ridge. *J. Geophys. Res. Oceans (Under Review)*, 2026.

[5] **Wenrui Jiang** and Thomas W. N. Haine. Generation and propagation of Eastern Subpolar North Atlantic salinity anomalies. *Journal of Geophysical Research: Oceans*, 131(10), September 2026.

[6] **Wenrui Jiang**, Gael Forget, Yuanyuan Song, and Thomas W. N. Haine. Coupled air-sea interaction drove and sustained the 2013–2016 North Pacific Marine Heatwave. *Nat. Commun.*, August 2026.

[7] **Wenrui Jiang** and Thomas W. N. Haine. Tracer budgets on Lagrangian trajectories. *J. Advances Model. Earth Sys.*, 17(9), September 2025.

[8] Thomas W. N. Haine, Stephen M. Griffies, Geoffrey Gebbie, and **Wenrui Jiang**. A Review of Green's Function Methods for Tracer Timescales and Pathways in Ocean Models. *J. Advances Model. Earth Sys.*, 17(7), July 2025.

[9] **Wenrui Jiang**, Thomas W. N. Haine, and Mattia Almansi. Seaduck: A Python Package for Eulerian and Lagrangian interpolation on ocean datasets. *Journal of Open Source Software*, 8(92):5967, December 2023.

[10] Thomas W. N. Haine, Ali H. Siddiqui, and **Wenrui Jiang**. Arctic Freshwater Impact on the Atlantic Meridional Overturning Circulation: Status and Prospects. *Phil. Trans. R. Soc. Lond. A*, 381(2262), October 2023.

[11] **Wenrui Jiang**, Liguang Wu, and Qingyuan Liu. High-Wind Drag Coefficient Based on the Tropical Cyclone Simulated With the WRF-LES Framework. *Frontiers in Earth Science*, 9, May 2021.

Conference Proceedings

[1] **Wenrui Jiang**. Lagrangian Tracer Budget Based on ECCO (talk). ECCO Annual Meeting, 2026.

[2] **Wenrui Jiang**, Gaël Forget, Yuanyuan Song, and Thomas W.N. Haine. Coupled air-sea interaction drove and sustained the 2013–2016 North Pacific Marine Heatwaves (poster). OSM, 2026.

[3] Thomas W.N. Haine, and **Wenrui Jiang** (presenting). Generation and propagation of Eastern Subpolar North Atlantic salinity anomalies (talk). OSM, 2026.

[4] **Wenrui Jiang**, Thomas W.N. Haine, Peter P. Sullivan, and Charles Meneveau. Dynamics of Monochromatic Inertia–Gravity Waves Crossing an upper-ocean Submesoscale Cold Filament (poster). AMS AOFD, 2026.

[5] **Wenrui Jiang**, Yuanyuan Song (presenting), Gaël Forget, and Thomas W.N. Haine. Coupled air-sea interaction drove and sustained the 2013–2016 North Pacific Marine Heatwaves (poster). AGU, 2025.

[6] Gaël Forget, **Wenrui Jiang**, Yuanyuan Song, and Thomas W.N. Haine. Diagnosing upper ocean heat anomaly with Lagrangian budget in ECCO (talk). ECCO summer school, 2025.

[7] Thomas W.N. Haine and **Wenrui Jiang**. Democratize the Data: A New Way to Analyze Ocean Models (townhall). Ocean Sciences Meeting 2024. AGU, 2024.

[8] **Wenrui Jiang** and Thomas W.N. Haine. Lagrangian budget on the subpolar north atlantic salinity anomalies (poster). Ocean Sciences Meeting 2024. AGU, 2024

Invited Talks

[1] Poseidon Viewer: a Visualization Tool for Petabyte-scale Ocean Data, Free and Open Source Software Project Fund (FOSSProF) Summative Event, Oct. 7th, 2024.

[2] Lagrangian Budget Description of Salinity Anomalies in the Eastern Subpolar North Atlantic, MIT SLS seminar, June 18th, 2024.

Open source projects

Seaduck *seaduck*

- A tool for interpolation, Lagrangian particle simulation, and closing tracer budget on Lagrangian trajectories.
- Develop, test, and continuously maintain.
- Documentation: macekuailv.github.io/seaduck/

Poseidon viewer *poseidon-viewer*

- A visualization tool for the LLC4320 ocean dataset.
- Develop the backend interpolator and the documentation.
- Documentation: sciserver.github.io/poseidon-viewer

Oceanspy *oceanspy*

- A Python package to facilitate ocean model data analysis and visualization maintained by Haine's group.
- Maintenance and updates.
- Documentation: oceanspy.readthedocs.io

Current Research Projects

Stochastic Lagrangian modeling of vertical diffusion in Labrador Sea convection
Collaborators: *Gaël Forget*

- Developed a hybrid stochastic particle method that switches between a Fokker–Planck solver and a numerical stochastic differential equation integrator depending on local diffusivity, addressing ill-conditioning in the vertical diffusion transform matrix.
- Conducting Lagrangian particle experiments in the Labrador Sea to quantify transit times between Labrador Sea Water and neighboring water masses.

Dynamics of Monochromatic Inertia–Gravity Waves Crossing a Submesoscale Cold Filament
Collaborators: *Thomas W.N. Haine, Peter P. Sullivan, Charles Meneveau*

- Simulated the evolution of upper-ocean submesoscale filaments using NCAR-LES, introducing compact inertia-gravity wave (IGW) trains at the filament's mature stage.
- Constructed along-filament-averaged energy budgets decomposed into mean/turbulent available potential energy and kinetic energy (MAPE, TAPE, MKE, TKE).
- Found that prograde IGWs enhance shear production and dissipation by increasing momentum flux near the surface at the filament center.
- Showed via wave tracing that surface restratification by secondary circulations creates a duct for wave energy to reach the surface filament.

Teaching Experience

Spring 2023	TA and guest lecturer, 'Oceans and Atmospheres', Johns Hopkins University
Fall 2022	TA, 'Introduction to Global Environmental Change', Johns Hopkins University

Mentoring

- Co-supervised Qibo Su (undergraduate, Fall 2025). "AmazonHydroViewer: Interactive Web Dashboard for Visualizing Ensemble Hydrometeorological Forecasts and Climatology Across the Amazon Basin." Presented at U.S. Senate.
- Co-supervised Jie Ma (master's capstone project, Fall 2023). "Transit Times and Pathways of Fukushima Nuclear Wastewater Release."

Awards

2025	Best Graduate Student Paper Award
2020	The 2nd Conditional Nonlinear Optimum Perturbation (CNOP) method summer school: Outstanding Student
2020	Interdisciplinary Contest in Modeling: Meritorious Winner
2018	First-class Scholarship (2nd out of 80)
2017	Freshman Scholarship (Top 5%)

Leadership and Volunteering

Sept. 2023 - June 2024	Website Manager for Chinese-American Oceanic and Atmospheric Association
Feb. 2023 - July 2023	Host of "Ocean Seminar" among Oceanographers in Maryland
Feb. 2023 - present	Host of "A&O Seminar/Meeting" at JHU, a weekly research exchange among graduate students and faculty
March 2021 - May 2021	Volunteer Teacher, Qingpu, China
Jan. 2019 - Aug. 2019	President of Fudan Astronomy Club

Additional Information

Languages: English (fluent), Mandarin Chinese (native), French (elementary to intermediate)